FIT1058 — Chapter 3: Proofs

Exercise Solutions

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Semester 2, 2026

Exercise 1

Recall the Collatz function

$$\text{Collatz}(x) = \begin{cases} 3x+1 & \text{if } x \text{ is odd} \\ \dfrac{x}{2} & \text{if } x \text{ is even} \end{cases}$$

Prove by cases that, for all positive integers n,

$$\text{Collatz}^{(2)}(n) \leq \tfrac{3}{2}n + 1.$$

Proof. Let $\text{Collatz}(x) = C(x)$.

Case 1: x is odd, $x = 2n+1$.

$$C(x) = 3(2n+1)+1 = 6n+4 = 2(3n+2) \quad \text{(even)}$$
$$C^{(2)}(x) = 3n+2 = 3\left(\frac{x-1}{2}\right) + 2 = \frac{3x-3}{2} + 2 = \frac{3}{2}x + \frac{1}{2}$$

$$\implies C^{(2)}(x) < \frac{3}{2}x + 1.$$

Case 2: x is even, $x = 2m$.

$$C(x) = m, \quad (m \text{ is odd})$$
$$C^{(2)}(x) = 3m+1 = 3\left(\frac{x}{2}\right) + 1 = \frac{3}{2}x + 1$$

$$\implies C^{(2)}(x) = \frac{3}{2}x + 1.$$

So, $C^{(2)}(x) \leq \dfrac{3}{2}x + 1$. ∎

Exercise 2

A set of strings L is called *hereditary* if it has the following property: for every nonempty string x in L, there is a character in x which can be deleted from x to give another string in L.

Prove by contradiction that every nonempty hereditary set of strings contains the empty string.

Proof (by contradiction). Assume that L is a nonempty hereditary set of strings that does *not* contain the empty string.

Let $S = \{\, |x| : x \in L \,\}$ be the set of lengths of strings in L. Since L is nonempty, S is a nonempty subset of $\mathbb{N}$. By the Well-Ordering Principle, S has a smallest element.

Let $x_0 \in L$ be a string of minimum length, i.e.

$$\forall x \in L,\ |x_0| \le |x|.$$

Since x_0 is a nonempty string, its length is $|x_0| = n_0 \ge 1$.

Since L is hereditary and $x_0 \in L$ is nonempty, there is a character in x_0 which can be deleted to give another string $x_1 \in L$, where

$$|x_1| = n_0 - 1.$$

Since x_0 is the smallest string in L, we have $|x_0| \le |x|$ for every $x \in L$. But

$$|x_1| = n_0 - 1 < n_0 = |x_0|,$$

which contradicts the minimality of x_0.

This is a contradiction. Therefore, the assumption is false, and L must contain the empty string. ∎

Exercise 5

Let E be a finite set and let $\mathcal{I}$ be a collection of subsets of E that satisfies the following three properties:

(I1) $\emptyset \in \mathcal{I}$.

(I2) Whenever $X \subseteq Y$ and $Y \in \mathcal{I}$, we also have $X \in \mathcal{I}$. This is called the *hereditary property*.

(I3) Whenever $X, Y \in \mathcal{I}$ and $|X| < |Y|$, there exists $y \in Y \setminus X$ such that $X \cup \{y\} \in \mathcal{I}$. In other words, given two different-sized sets in $\mathcal{I}$, you can always enlarge the smaller set, using some new member from the larger set, to get another set in $\mathcal{I}$. This is called the *independence augmentation property*.

The sets in $\mathcal{I}$ are called *independent*.

For example, if $E = \{a, b, c\}$, then each of the following collections satisfies all three properties (I1)–(I3):

- $\mathcal{I} = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$
- $\mathcal{I} = \{\emptyset, \{a\}, \{b\}, \{c\}\}$
- $\mathcal{I} = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, c\}, \{b, c\}\}$
- $\mathcal{I} = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}\}$
- $\mathcal{I} = \{\emptyset\}$

By way of counterexample, using the same E, each of the following collections fails to satisfy all three properties (I1)–(I3):

- $\mathcal{I} = \{\{a\}, \{b\}, \{a, b\}\}$. This fails (I1) and (I2).
- $\mathcal{I} = \{\emptyset, \{a\}, \{a, b\}\}$. This fails (I2).

- $\mathcal{I} = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}\}$. This satisfies (I1) and (I2) but fails (I3).
- $\mathcal{I} = \emptyset$. This fails (I1).

(a) For each counterexample above that fails (I2) and/or (I3), give specific sets X and Y for which the property fails.

(b) Prove that, for a social network, the set of cliques satisfies (I1) and (I2) but not, in general, (I3).

(c) Prove by contradiction that all maximal independent sets have the same size.

Suppose that each member $e \in E$ has a nonnegative real weight $w(e)$, so $w : E \to \mathbb{R}_0^+$. We use this weight function to also give weights to subsets of E, by defining the weight of a set to be the sum of the weights of its elements: for all $X \subseteq E$, define

$$w(X) := \sum_{e \in X} w(e).$$

Suppose we seek an independent set of maximum weight. Consider the following *greedy algorithm*, so called because it always makes the choice that gives biggest immediate gain:

Algorithm 1: Greedy algorithm for maximum weight independent set

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Input: a finite set E, a function $w : E \to \mathbb{R}_0^+$, a collection $\mathcal{I}$ of subsets of E
Output: an independent set X

$X \leftarrow \emptyset$
$i \leftarrow 1$
while $\exists\, y \in E \setminus X$ *such that* $X \cup \{y\} \in \mathcal{I}$ **do**
  choose y with largest weight $w(y)$ among all such candidates
  $e_i \leftarrow y$
  $X \leftarrow X \cup \{e_i\}$
  $i \leftarrow i + 1$
return X

(d) Construct a simple social network, and give a nonnegative real weight to each person, such that the greedy algorithm, with $\mathcal{I}$ being the set of all cliques, does not find the maximum weight clique.

Now suppose $\mathcal{I}$ satisfies (I1)–(I3). The greedy algorithm chooses elements of E in a specific order, represented as $e_1, e_2, \ldots, e_r$.

(e) Prove by contradiction that the weights of the chosen elements are decreasing, i.e.,

$$w(e_1) \geq w(e_2) \geq \cdots \geq w(e_r).$$

("Decreasing" allows the possibility that some weights stay the same; we don't require strictly decreasing.)

(f) Prove that the greedy algorithm finds a maximum weight independent set.

Solution. (a)

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Solution. (b)

Proof of (b). Let E = vertices of a social network, and let $\mathcal{I}$ = the set of all cliques.

(I1). Let X be a clique, and consider $X = \emptyset$. Since $\emptyset$ contains no elements, there do not exist $u, v \in \emptyset$ with $u \neq v$. Hence the implication "$u \neq v \implies (u, v)$ is edge" holds vacuously for all $u, v \in \emptyset$. So $\emptyset$ is a clique, i.e. $\emptyset \in \mathcal{I}$.

(I2). Need to prove: if $X \subseteq Y$ and Y is a clique, then X is a clique.

For all $u, v \in X$ with $u \neq v$: since $X \subseteq Y$, we have $u, v \in Y$ with $u \neq v$. Since Y is a clique, (u, v) is an edge. Hence X is also a clique.

(I3) fails in general. Consider the graph with vertices $\{a, b, c, d\}$ and a single edge (b, c). Let $X = \{a\}$ and $Y = \{b, c\}$.

X and Y are cliques. $Y \setminus X = \{b, c\}$:

- $X \cup \{b\} = \{a, b\}$ is not a clique, since (a, b) is not an edge.
- $X \cup \{c\} = \{a, c\}$ is not a clique, since (a, c) is not an edge.

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Solution. (c)

Proof of (c). An independent set $X \in \mathcal{I}$ is *maximal* if $\nexists Y \in \mathcal{I}$ such that $X \subsetneq Y$.

We need to prove: $\forall X, Y \in \mathcal{I}$, if X maximal and Y maximal, then $|X| = |Y|$.

Assume, for contradiction, there exist two maximal independent sets $X, Y \in \mathcal{I}$ with $|X| \neq |Y|$. WLOG, assume $|X| < |Y|$.

Since $X, Y \in \mathcal{I}$ and $|X| < |Y|$, there exists $y \in Y \setminus X$ such that $X \cup \{y\} \in \mathcal{I}$.

Since $y \notin X$, we have $X \subsetneq X \cup \{y\}$, and $X \cup \{y\} \in \mathcal{I}$. This contradicts the assumption that X is maximal.

Therefore, the assumption is false: all maximal independent sets have the same size. ■

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Solution. (d)

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Solution. (e)

Proof of (e). When $\mathcal{I}$ satisfies (I1)–(I3), the greedy algorithm chooses elements of E in a specific order $e_1, e_2, \ldots, e_r$. We prove $w(e_i) \geq w(e_{i+1})$ for all $1 \leq i < r$.

Note that e_i is the maximum-weight element among all $y \in E \setminus X_{i-1}$ satisfying $X_{i-1} \cup \{y\} \in \mathcal{I}$, i.e.

$$X_{i-1} \cup \{e_i\} \in \mathcal{I}, \qquad \forall y \text{ with } X_{i-1} \cup \{y\} \in \mathcal{I},\ w(e_i) \geq w(y).$$

Assume, for contradiction, that $w(e_i) < w(e_{i+1})$ for some $1 \leq i < r$.

Since e_{i+1} was chosen by the algorithm at round $i + 1$,

$$X_i \cup \{e_{i+1}\} \in \mathcal{I}.$$

Since $X_{i-1} \subseteq X_i$, we have $X_{i-1} \cup \{e_{i+1}\} \subseteq X_i \cup \{e_{i+1}\}$. Since $X_i \cup \{e_{i+1}\} \in \mathcal{I}$, by (I2),

$$X_{i-1} \cup \{e_{i+1}\} \in \mathcal{I}.$$

This means e_{i+1} is one of the candidates y satisfying $X_{i-1} \cup \{y\} \in \mathcal{I}$ at round i. Since e_i was chosen at round i as the maximum-weight element among all such candidates, we must have

$$w(e_i) \geq w(e_{i+1}),$$

which contradicts the assumption $w(e_i) < w(e_{i+1})$.

Therefore, the assumption is false, and $w(e_i) \geq w(e_{i+1})$ for all $1 \leq i < r$. ■

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Solution. (f)

Proof of (f). Let Y be a maximum weight independent set. By the stopping condition of the greedy algorithm, when $X = \{e_1, \ldots, e_r\}$, there is no $y \in E \setminus X$ such that $X \cup \{y\} \in \mathcal{I}$. So X is a maximal independent set.

By part (c), we know $|X| = |Y| = r$. Write $Y = \{f_1, \ldots, f_r\}$ with $w(f_i) \geq w(f_{i+1})$ for $1 \leq i \leq r$.

We need to prove that for each $i = 1, \ldots, r$, $w(e_i) \geq w(f_i)$.

By contradiction (minimal counterexample): assume there exists a minimum i such that $w(e_i) < w(f_i)$. By minimality of i, we have $w(e_k) \geq w(f_k)$ for all $k < i$.

Let $X_{i-1} = \{e_1, \ldots, e_{i-1}\}$ and $F_i = \{f_1, \ldots, f_i\}$. Since $X_{i-1} \subseteq X \in \mathcal{I}$, by (I2), $X_{i-1} \in \mathcal{I}$. Since $F_i \subseteq Y \in \mathcal{I}$, by (I2), $F_i \in \mathcal{I}$. Also $|X_{i-1}| = i - 1 < i = |F_i|$.

By (I3), there exists $y \in F_i \setminus X_{i-1}$ such that $X_{i-1} \cup \{y\} \in \mathcal{I}$.

Since $y \in F_i = \{f_1, \ldots, f_i\}$, and these are sorted in decreasing order of weight, f_i has the smallest weight among them, so

$$w(y) \geq w(f_i).$$

Now, since $y \notin X_{i-1}$ and $X_{i-1} \cup \{y\} \in \mathcal{I}$, y is a valid candidate at round i of the greedy algorithm. But e_i was chosen at round i as the element of maximum weight among all such valid candidates, so

$$w(e_i) \geq w(y) \geq w(f_i).$$

This contradicts the assumption $w(e_i) < w(f_i)$. Hence no such i exists, and $w(e_i) \geq w(f_i)$ for all $i = 1, \ldots, r$.

Summing over i:

$$w(X) = \sum_{i=1}^{r} w(e_i) \geq \sum_{i=1}^{r} w(f_i) = w(Y).$$

Since Y was assumed to be a maximum weight independent set, $w(X) \geq w(Y)$ forces $w(X) = w(Y)$. Therefore X is also a maximum weight independent set.

$$\therefore \text{ Greedy algorithm finds a maximum weight independent set.}$$

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Exercise 10

Consider the following algorithm:

Algorithm 2: Greedy algorithm for maximum weight independent set

Input: a finite set E, a function $w : E \to \mathbb{R}_0^+$, a collection $\mathcal{I}$ of subsets of E

Output: an independent set X

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X ← ∅
i ← 1
while ∃y ∈ E \ X such that X ∪ {y} ∈ I do
    choose y with largest weight w(y) among all such candidates
    e_i ← y
    X ← X ∪ {e_i}
    i ← i + 1
return X
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(a) For each pair i, j, how many times is the action inside the innermost loop performed? Express this in terms of i or j or both.

(b) For each i, how many times is the action inside the innermost loop performed? Write this as both a sum of some number of terms, and (using a theorem in this chapter) a simple algebraic expression, both in terms of i.

(c) Now write a sum of some number of terms, in terms of n, for the total number of times the action is performed.

(d) Work out this expression for $n = 1, 2, 3, 4, 5$, and for as many further values of n as you like.

(e) Explore how these numbers behave.

(f) Conjecture a simple algebraic expression, in terms of n, for the total number of times the action is performed.

(g) Prove, by induction on n, that your expression is correct.

Solution.

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Exercise 11

The n-th harmonic number H_n is defined by

$$H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}.$$

In this exercise we prove by induction that $H_n \geq \ln(n+1)$. (It follows that the harmonic numbers increase without bound, even though H_n grows more and more slowly as n increases.)

(i) **Inductive basis:** prove that $H_1 \geq \ln(1+1)$.

(ii) Let $n \geq 1$. Assume $H_n \geq \ln(n+1)$ (inductive hypothesis). Consider H_{n+1}; write it recursively using H_n. Use the inductive hypothesis to obtain $H_{n+1} \geq \ldots$, then use the fact that $\ln(1+x) \leq x$, and a property of logarithms, to show $H_{n+1} \geq \ln(n+2)$.

(iii) What can you now conclude, and why?

Advanced afterthoughts.

- The inequality implies $H_n \geq \ln n$, since $\ln(n+1) \geq \ln n$. It is instructive to try to prove this directly by induction — you will probably run into a snag, illustrating that induction sometimes requires proving something stronger than what you set out to prove.
- Would your proof work for logarithms to other bases, apart from e? Where does the proof use base e?
- It is known that $H_n \leq (\ln n) + 1$. Can you prove this?

Solution.

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Exercise 12

(Challenge) Let E be a finite set, and let $\mathcal{S}$ be a set of subsets of E. We say $\mathcal{S}$ is *linear under* $\triangle$ if, for every two sets $X, Y \in \mathcal{S}$, we have $X \triangle Y \in \mathcal{S}$.

The *span* of $\mathcal{S}$, written $\text{span}(\mathcal{S})$, is the set of all symmetric differences of any number of members of $\mathcal{S}$:

$$\text{span}(\mathcal{S}) := \{X_1 \triangle X_2 \triangle \cdots \triangle X_k : X_1, \ldots, X_k \in \mathcal{S},\ k \in \mathbb{N}\}.$$

By convention, the symmetric difference of zero sets is $\emptyset$, so $\text{span}(\mathcal{S})$ always contains $\emptyset$.

$\mathcal{S}$ is *dependent under* $\triangle$ if, for some distinct $X_1, \ldots, X_k \in \mathcal{S}$ (i.e. $X_i \neq X_j$ for $i \neq j$), $X_1 \triangle \cdots \triangle X_k = \emptyset$. If $\mathcal{S}$ is not dependent under $\triangle$, it is *independent under* $\triangle$.

$\mathcal{S}$ is a *circuit under* $\triangle$ if it is dependent under $\triangle$ and also minimal with respect to that property. $\mathcal{S}$ is a *base under* $\triangle$ if it is independent under $\triangle$ and also maximal with respect to that property.

(a) Let $\mathcal{S}$ be linear under $\triangle$. Prove, by induction on k, that for all k and all $X_1, \ldots, X_k \in \mathcal{S}$, $X_1 \triangle \cdots \triangle X_k \in \mathcal{S}$.

(b) Prove that, for every set $\mathcal{S}$ of subsets of E, $\mathcal{S}$ is linear under $\triangle$ if and only if $\mathcal{S} = \text{span}(\mathcal{S})$.

(c) Prove by induction on k that for all k and all $X_1, \ldots, X_k \in \mathcal{S}$, either $X_1 \triangle \cdots \triangle X_k = \emptyset$, or there exist distinct $Y_1, \ldots, Y_n \in \mathcal{S}$ with $n \leq k$ such that $X_1 \triangle \cdots \triangle X_k = Y_1 \triangle \cdots \triangle Y_n$.

(d) Prove that $\mathcal{S}$ is dependent under $\triangle$ if and only if there exists $X \in \mathcal{S}$ such that $X \in \text{span}(\mathcal{S} \setminus \{X\})$.

(e) Prove that if $\mathcal{S}$ is a minimal dependent set under $\triangle$, then for every $X \in \mathcal{S}$ we have $X \in \text{span}(\mathcal{S} \setminus \{X\})$.

(f) Prove that every minimal spanning set under $\triangle$ is independent under $\triangle$.

(g) Prove that every minimal spanning set under $\triangle$ is a maximal independent set under $\triangle$.

(h) Prove that every maximal independent set under $\triangle$ is also a spanning set under $\triangle$.

(i) Prove that every maximal independent set under $\triangle$ is also a minimal spanning set under $\triangle$.

(j) Prove that $\mathcal{S}$ is a minimal spanning set under $\triangle$ if and only if it is a maximal independent set under $\triangle$.

(k) Prove that, if $\mathcal{S}$ is independent under $\triangle$ then $|\text{span}(\mathcal{S})| = 2^{|\mathcal{S}|}$.

(l) Prove the independence augmentation property: if $\mathcal{S}$ and $\mathcal{T}$ are independent under $\triangle$ and $|\mathcal{T}| = |\mathcal{S}| + 1$, then there exists $X \in \mathcal{T} \setminus \mathcal{S}$ such that $\mathcal{S} \cup \{X\}$ is independent under $\triangle$.

(m) Prove that all bases under $\triangle$ have the same size.

(n) Do all minimal dependent sets under $\triangle$ have the same size? Give a proof of your claim.

(o) Prove that, for every base $\mathcal{B}$ of $\mathcal{S}$ under $\triangle$, and every $X \in \mathcal{S}$, there is a unique way to write X as a symmetric difference of members of $\mathcal{B}$, using each element of $\mathcal{B}$ at most once.

For each previous part (a)–(o), state what type of proof you have used.

Finally: design a simple algorithm for the following problem — given a collection $\mathcal{S}$ of subsets of E with a nonnegative real weight for each member, output a subset $\mathcal{X}$ of $\mathcal{S}$ that is independent under $\triangle$ such that the sum of the weights of all members of $\mathcal{X}$ is maximum. Does this algorithm always find the best possible solution? Justify your answer, using a previous exercise.

Solution. (a) ■

Solution. (b) ■

Solution. (c) ■

Solution. (d) ■

Solution. (e) ■

Solution. (f) ■

Solution. (g) ■

Solution. (h) ■

Solution. (i) ■

Solution. (j) ■

Solution. (k) ■

Solution. (l) ■

Solution. (m) ■

Solution. (n) ■

Solution. (o) ■

Solution. Algorithm & justification

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Exercise 13

Identify the errors in the following "theorem" and "proof", which are both incorrect. The steps in the "proof" are numbered for convenience.

"Theorem." For all $n \in \mathbb{N}$, a circular disc can be cut into 2^n pieces by n straight lines.
"Proof."

1. Consider $n = 0$. The disc is not cut at all, so it is still in one piece. So the number of pieces is $2^0 = 1$. So the claim is true for $n = 0$.

2. Consider $n = 1$. A single line across a disc cuts it into two pieces, so the number of pieces is 2^1. So the claim is true for $n = 1$.

3. Now consider $n = 2$. If the circle is cut by one line into two pieces, then we can cut it again by a line that crosses the first line. Such a line must cut right across both of the pieces created by the first cut. Therefore each of those pieces is divided in two, so the total number of pieces is $2 \times 2 = 2^2$, so the claim is true for $n = 2$.

4. It is clear from the cases considered so far that, once the circle is cut by some number of lines, then another line can be used to divide every piece into two pieces, thereby doubling the number of pieces.

5. So, if we use n lines, then the repeated doubling gives 2^n pieces. Therefore the claim is true for all n. "□"

Solution.

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Exercise 14

Identify the errors in the following "theorem" and "proof", which are both incorrect.

"Theorem." Every rational number is an integer.
"Proof."

1. We need to show that the sets $\mathbb{Z}$ and $\mathbb{Q}$ are equal.

2. To prove that two sets are equal, we can prove the appropriate subset and superset relations between the two sets.

3. We first prove the subset relation between these two sets, $\mathbb{Z} \subseteq \mathbb{Q}$.

4. Let $x \in \mathbb{Z}$.

5. Since x is an integer, we have $x = \dfrac{x}{1}$.

6. So x is a quotient of integers.

7. Therefore x is rational, i.e., $x \in \mathbb{Q}$.

8. So we have proved that $\mathbb{Z} \subseteq \mathbb{Q}$.

9. Now we need to prove the superset relation, which is the reverse of subset.
10. So we will prove that $\mathbb{Q} \supseteq \mathbb{Z}$.
11. Let $x \notin \mathbb{Q}$.
12. Then x cannot be written in the form $\dfrac{p}{q}$ where p and q are integers.
13. So it certainly cannot be written in this form with $q = 1$.
14. Therefore x is not an integer, i.e., $x \notin \mathbb{Z}$.
15. So we have proved that $\mathbb{Q} \supseteq \mathbb{Z}$.
16. It follows from our subset and superset relations between the two sets that $\mathbb{Z} = \mathbb{Q}$. "□"

"God made the integers; all else is the work of man." — Leopold Kronecker (1823–1891)

Solution.

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Exercise 15

Let G be the cryptosystem obtained from Caesar slide by restricting the keyspace to the first half of the alphabet (see Exercise 2.15 for definitions relating to the Caesar slide cryptosystem). So the keyspace K_G is

$$\{\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{d}, \mathrm{e}, \mathrm{f}, \mathrm{g}, \mathrm{h}, \mathrm{i}, \mathrm{j}, \mathrm{k}, \mathrm{l}, \mathrm{m}\}.$$

Prove that G is not idempotent.

Solution.

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